

The number of normal measures, revisited

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Abstract

We present a new version of the Friedman–Magidor theorem: for every measurable cardinal κ and $\tau \leq \kappa^{++}$, there exists a forcing extension $V \subseteq V[G]$ such that any normal measure $U \in V$ on κ has exactly τ distinct lifts in $V[G]$, and every normal measure on κ in $V[G]$ arises as such a lift. This version differs from the original Friedman–Magidor theorem in several notable ways. First, the new technique does not involve forcing over canonical inner models or rely on any fine-structural tools or assumptions, allowing it to be applied in the realm of large cardinals beyond the current reach of the inner model program. Second, in the case where $\tau \leq \kappa^+$, all lifts of a normal measure $U \in V$ on κ to $V[G]$ have the same ultrapower. Finally, the technique generalizes to a version of the Friedman–Magidor theorem for extenders. An additional advantage is that the forcing used is notably simple, relying only on nonstationary support product forcing.

The analysis of measures in forcing extensions plays a central role in the theory of large cardinals. A major question in this area was whether the existence of a model of ZFC with exactly two normal measures follows from the consistency of ZFC with a measurable cardinal. This question was answered affirmatively by the following landmark theorem of Friedman and Magidor.

Theorem 0.1. (*Friedman–Magidor theorem, [9]*) *Suppose $V = L[U]$ is the minimal inner model in which there is a measurable cardinal κ , and let $\tau \leq \kappa^{++}$. Then in a cardinal-preserving forcing extension, κ is measurable and carries exactly τ normal measures.*

In their work, Friedman and Magidor identified the various components that contribute to the construction of normal measures in forcing extensions. Their forcing notion includes, for each such component, a specialized feature designed to control and constrain it. This naturally led to the orchestration of several forcing techniques, each playing an essential role in the proof. Indeed, Theorem 0.1 stands out for the range of methods it brings together, including forcing over canonical inner models, nonstationary support iterations, generalized Sacks forcing, and interleaved self-coding posets. These techniques have since enabled a variety of further applications, including [4], [3], [16], [2], and [5].

The central result of this paper is a significantly simpler and more transparent proof of the Friedman–Magidor theorem. Our approach dispenses entirely with inner model theory, self-coding, and generalized Sacks forcing. Instead, it relies solely on a product forcing with nonstationary support.

Similarly to the original Friedman–Magidor proof, we concentrate on the case where the expected number τ of normal measures is at most κ^+ ; achieving $\tau = \kappa^{++}$ is done by performing the Kunen–Paris forcing (see [14]).

The main theorem of this paper is the following:

Theorem 0.2. *Suppose that κ is a measurable cardinal and $\tau \leq \kappa^+$. There exists a poset $\mathbb{R}^\tau$ which preserves κ, κ^+ and the measurability of κ , such that, whenever $G \subseteq \mathbb{R}^\tau$ is generic over V , the following properties hold:*

1. *Every normal measure $U \in V$ on κ has exactly τ distinct lifts $\langle U_\eta^* : \eta < \tau \rangle$ to normal measures on κ in $V[G]$, all of which have the same ultrapower over $V[G]$.*
2. *Every normal measure $W \in V[G]$ on κ has the form U_η^* for some normal $U \in V$ and $\eta < \tau$.*

Also, assuming GCH in V , $\mathbb{R}^\tau$ preserves cardinals.

In the case where $\tau < \kappa$, $\mathbb{R}^\tau$ has the form $\text{Cohen}(\omega) * \mathbb{P}^\tau$, where $\mathbb{P}^\tau$ consists of partial functions $f : \kappa \rightarrow \tau$ whose support is nowhere stationary, ordered by inclusion.¹ In the case where $\tau \in \{\kappa, \kappa^+\}$, $\mathbb{P}^\tau$ is slightly modified using canonical functions, see definition 2.1 below.

As mentioned above, one of the main advantages of Theorem 0.2 is that it does not involve forcing over canonical inner models. In the original Friedman–Magidor proof, forcing over $L[U]$ played a significant role in two ways. First, by assuming that $L[U]$ is the core model and forcing over it, every normal measure on κ in the generic extension restricts to an iterated ultrapower of $L[U]$. (This is the starting point of a sequence of classical results in inner model theory, culminating in a well-known theorem of Schindler, see [17]). Second, canonical inner models possess fine structure, which Friedman and Magidor utilized by incorporating self-coding posets into their forcing construction. (See [9] for further details on self-coding and its role in constraining the number of normal measures).

While these techniques demonstrate the utility of inner model theory and fine structure in forcing theory, forcing over canonical inner models can be restrictive—particularly because it does not apply at levels of the large cardinal hierarchy not yet reached by the inner model program. Since our proof of Theorem 0.2 avoids the use of inner model theory, it can be employed to obtain consistency results at the level of strongly compact and supercompact cardinals, such as the following:

¹The purpose of the initial Cohen forcing is to ensure that $\mathbb{R}^\tau$ has a gap below κ (see Theorem 2.1).

Theorem 0.3 (Apter, K., Poveda). *Assume the consistency of a supercompact cardinal and the linearity of the Mitchell order.*

1. *It is consistent that there exists a supercompact, and there are exactly two normal measures on the least measurable cardinal.*
2. *Assume in addition a measurable cardinal above the supercompact. Then it is consistent that there is a supercompact cardinal, and the least measurable cardinal above it carries exactly two normal measures.*

Theorem 0.3 and its proof are expected to appear in a joint work with Apter and Poveda. The first clause follows directly from Theorem 0.2 and the fact that the forcing $\mathbb{R}^\tau$ (for $\tau = 2$) is small relative to the least supercompact cardinal above κ . The second is obtained by forcing with a variation of $\mathbb{P}^\tau$ after the universe has been suitably prepared so that the supercompactness of κ is indestructible under κ -directed-closed forcing.

Our forcing notions $\mathbb{P}^\tau$ (for $\tau < \kappa$) have several additional applications. In Section 3, we classify all possible lifts of U^n to the generic extension obtained by forcing with $\mathbb{P}^\tau$, where $U \in V$ is a normal measure on κ and $n < \omega$. When this analysis is carried out in the case where $V = L[U]$ is the core model, it allows for a complete description of the σ -complete ultrafilters in the extension. This yields a ZFC model with an interesting measure structure:

Theorem 0.4. *Assume that $V = L[U]$, and let κ be the measurable cardinal. Fix $\tau < \kappa$. Let $G \subseteq \mathbb{P}^\tau$ be generic over V . Then in $L[U][G]$:*

1. *there are τ normal measures on κ , $\langle U_\eta^* : \eta < \tau \rangle$, all of them have the same ultrapower.*
2. *for every σ -complete ultrafilter $W \in L[U][G]$, the ultrapower $\text{Ult}(L[U][G], W)$ is a finite iterated ultrapower of $L[U][G]$ by normal measures.*
3. *every ultrapower of $L[U][G]$ (via a σ -complete ultrafilter) is isomorphic to $\text{Ult}(L[U][G], (U_0^*)^n)$ for some $n < \omega$.*

Theorem 0.4 is related to the Weak Ultrapower Axiom. Recall that Goldberg's Ultrapower Axiom (UA) states that for any pair of σ -complete ultrafilters U, W , there are σ -complete ultrafilters $U^* \in M_W$ and $W^* \in M_U$,² such that $M_{U^*}^{M_W} = M_{W^*}^{M_U}$ and $j_{W^*} \circ j_U = j_{U^*} \circ j_W$. The Weak Ultrapower Axiom is obtained from UA by removing the demand that $j_{W^*} \circ j_U = j_{U^*} \circ j_W$. Theorem 0.4 (for $1 < \tau < \kappa$) demonstrates that a model of the Weak UA and the negation of UA³ can be constructed assuming the consistency of a measurable cardinal. This slightly improves a previous result by Goldberg and Benhamou (see [7]), who constructed such a model assuming the consistency of a measurable limit of measurables.

²Here and below, for a σ -complete ultrafilter U , we denote by M_U the transitive collapse of $\text{Ult}(V, U)$, and by $j_U : V \rightarrow M_U$ the corresponding ultrapower embedding.

³The negation of UA holds in the model of Theorem 0.4 since, by a theorem of Goldberg, UA implies the linearity of the Mitchell order. See [11] for more details.

Corollary 0.1. *Weak $UA + \neg UA$ is equiconsistent with a single measurable cardinal.*

Finally, building on the techniques introduced earlier, we generalize the Friedman–Magidor framework from normal measures to extenders in Theorem 4.1. A typical application of this theorem is illustrated by the following example:

Theorem 0.5. *Assume GCH and let κ be a $(\kappa + 2)$ -strong-cardinal. Then for every $\tau \leq \kappa^{++}$, there exists a cardinal-preserving forcing extension $V[G]$ in which every (κ, κ^{++}) -extender E that witnesses κ being $(\kappa + 2)$ -strong, lifts to exactly τ nonequivalent⁴ (κ, κ^{++}) -extenders of $V[G]$, and all of them have the same ultrapower and witness $(\kappa + 2)$ -strongness over $V[G]$. Moreover, every (κ, κ^{++}) -extender witnessing that κ is $(\kappa + 2)$ -strong in $V[G]$ arises as such a lift.*

Goldberg observed that the bound κ^{++} on the number of (κ, κ^{++}) -extenders having the same ultrapower is optimal under GCH (see Theorem 4.3). Similar results can be obtained for extenders with higher degrees of strongness.

The structure of this paper is as follows: In Section 1 we cover the basic properties of nonstationary support products. In Section 2 we prove Theorem 0.2. In Section 3 we discuss lifting iterates of U^n for a normal measure U and $n < \omega$, and prove Theorem 0.4. In Section 4 we phrase and prove a Friedman–Magidor theorem for extenders (Theorem 4.1), and apply it to prove Theorem 0.5. In Section 5 we discuss several open questions.

Throughout the paper, we use the following convention in forcing: if a condition p extends a condition q , we write $q \leq p$, and the weakest condition in a forcing notion $\mathbb{P}$ is denoted by $0_{\mathbb{P}}$. The rest of the notation is standard.

Acknowledgments

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The author would like to thank Moti Gitik for pointing out the important correction that $\mathbb{P}^\tau$ should be replaced by $\mathbb{R}^\tau$ in Theorem 0.2.

⁴Two extenders are equivalent if they induce the same ultrapower embedding

1 Nonstationary support products

In this section, we briefly develop the basic framework of product forcing with a nonstationary support. All the ideas already appeared in one form or another in [9]. Our presentation is influenced from [6] and [1].

Definition 1.1. *A set of ordinals A is nowhere stationary⁵ if for every inaccessible cardinal α , $A \cap \alpha$ is nonstationary in α .*

Definition 1.2. *(Products with nonstationary support) Fix a cardinal κ and let $I \subseteq \kappa$ be an unbounded set of ordinals. For every $\alpha \in I$, let $\mathbb{Q}_\alpha$ be a forcing notion. The **nonstationary support product** $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ consists of conditions p which are functions with domain I , such that for every $\alpha \in I$, $p(\alpha) \in \mathbb{Q}_\alpha$, and the set*

$$\text{supp}(p) = \{\alpha \in I : p(\alpha) \neq 0_{\mathbb{Q}_\alpha}\}$$

is nowhere stationary.

The ordering is defined by setting that p extends q if for every $\alpha \in \text{supp}(q)$, $p(\alpha)$ extends $q(\alpha)$ in $\mathbb{Q}_\alpha$ (in particular, $\text{supp}(q) \subseteq \text{supp}(p)$).

Assuming that $\beta < \kappa$ and $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ is a nonstationary support product, $\mathbb{P}$ can be naturally factored to the form $\mathbb{P} \restriction \beta \times \mathbb{P} \restriction \beta$, where $\mathbb{P} \restriction \beta = \prod_{\alpha \in \beta \cap I}^{NS} \mathbb{Q}_\alpha$ and $\mathbb{P} \restriction \beta = \prod_{\alpha \in I \setminus \beta}^{NS} \mathbb{Q}_\alpha$, by identifying a condition $p \in \mathbb{P}$ with the pair $(p \restriction (I \cap \beta), p \restriction (I \setminus \beta))$. We denote in this case $p \restriction \beta = p \restriction (I \cap \beta)$ and $p \restriction \beta = p \restriction (I \setminus \beta)$.

The most important feature of nonstationary support products or iterations is their fusion property.

Lemma 1.3 (The Fusion Lemma). *Let κ be a limit of inaccessible cardinals, and let $I \subseteq \kappa$ be an unbounded set of inaccessibles below κ . Let $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ be a nonstationary support product. Assume that:*

1. *for every $\alpha \in I$, $\text{rank}(\mathbb{Q}_\alpha) < \min(I \setminus \alpha + 1)$.*
2. *for every $\alpha \in I$, $\mathbb{Q}_\alpha$ is α -closed.*

*Then $\mathbb{P}$ satisfies the **Fusion Property**; that is, given:*

- *a condition $p \in \mathbb{P}$,*
- *a sequence $\langle d(\alpha) : \alpha < \kappa \rangle$ of dense-open subsets of $\mathbb{P}$,*

there exists $p^ \geq p$ and a club $C \subseteq \kappa$ (if $\text{cof}(\kappa) = \omega$, C is an unbounded cofinal ω -sequence) such that, for every $\alpha \in C$, the set*

$$\{r \in \mathbb{P} \restriction \alpha + 1 : r \restriction p^* \restriction \alpha + 1 \in d(\alpha)\}$$

is dense in $\mathbb{P} \restriction \alpha + 1$ above $p^ \restriction \alpha + 1$.*

⁵We will adopt this terminology, although the phrase *nonstationary in inaccessibles* would be more accurate in this context. For our purposes, this distinction is irrelevant, since the nowhere stationary sets we consider will consist solely of inaccessible cardinals - for such sets, being nonstationary in inaccessibles implies being nonstationary in every ordinal of uncountable cofinality.

Proof. We first take care of the case where κ is singular. Fix an inaccessible $\nu \in (\text{cof}(\kappa), \kappa)$ and a continuous, cofinal, increasing sequence $\langle \nu_i : i < \text{cof}(\kappa) \rangle$ in κ , such that $\nu_0 > \nu$. Since $\mathbb{P} \restriction (\nu + 1)$ is more than $\max(\text{cof}(\kappa), |\mathbb{P} \restriction (\nu + 1)|)$ -closed, we may find a condition $p^* \geq p$ such that for every $i < \text{cof}(\kappa)$ and $r \in \mathbb{P} \restriction \nu + 1$, there exists $r' \geq r$ such that $r' \frown (p^* \restriction (\nu + 1)) \in d(\nu_i)$. It follows that for every $i < \text{cof}(\kappa)$,

$$\{r \in \mathbb{P} \restriction \nu_i + 1 : r \frown p^* \restriction (\nu_i + 1) \in d(\nu_i)\}$$

is dense in $\mathbb{P} \restriction (\nu_i + 1)$ above $p^* \restriction (\nu_i + 1)$. Thus, p^* and $C = \{\nu_i : i < \text{cof}(\kappa)\}$ are as desired.

Thus, assume from now on that κ is regular. We construct by induction sequences $\langle p_i : i < \kappa \rangle \subseteq \mathbb{P}$, $\langle \nu_i : i < \kappa \rangle \subseteq \kappa$ such that $\langle p_i : i < \kappa \rangle$ is a fusion sequence with respect to $\langle \nu_i : i < \kappa \rangle$, in the sense that the following properties are fulfilled:

1. $\langle \nu_i : i < \kappa \rangle$ is a continuous, cofinal sequence in κ .
2. $\langle p_i : i < \kappa \rangle$ is an increasing sequence of conditions extending p , such that, for every $i \leq j < \kappa$,

$$(a) \quad p_j \restriction (\nu_i + 1) = p_i \restriction (\nu_i + 1).$$

$$(b) \quad \nu_j \notin \text{supp}(p_i).$$

3. For every $i < \kappa$,

$$\{r \in \mathbb{P} \restriction \nu_i + 1 : r \frown (p_{i+1} \restriction (\nu_i + 1)) \in d(\nu_i)\}$$

is a dense-open subset of $\mathbb{P} \restriction \nu_i + 1$.

Assuming that such sequences have been constructed, we can set $p^* = \bigcup_{i < \kappa} p_i \restriction (\nu_i + 1)$. Since $C = \{\nu_i : i < \kappa\}$ is a club disjoint from $\text{supp}(p^*)$, p^* is a legitimate condition in $\mathbb{P}$.⁶ It's then not hard to verify that p^* and C are as required.

Thus, it suffices to prove that sequences $\langle p_i : i < \kappa \rangle$ and $\langle \nu_i : i < \kappa \rangle$ with the desired properties can be constructed. We define these sequences inductively, along with an auxiliary sequence $\langle C_i : i < \kappa \rangle$ of clubs in κ , such that, in addition to the above requirements,

4. for every $i < \kappa$, C_i is disjoint from $\text{supp}(p_i)$.
5. for every $i < j < \kappa$, $\nu_j \in C_i$ and $C_j \subseteq C_i$.

We proceed to the construction. For $i = 0$, take $p_0 = p$, $\nu_0 = \nu$ and C_0 any club disjoint from $\text{supp}(p_0)$.

For the successor step, assume that p_i, ν_i, C_i have been constructed. First set $\nu_{i+1} = \min(C_i \setminus (\nu_i + 1))$. Define $p_{i+1} \geq p_i$ such that:

⁶Note that we require that $\text{supp}(p^*)$ is nonstationary in any inaccessible $\lambda < \kappa$ as well; this follows immediately since given such λ , $p^* \restriction \lambda = p_i \restriction \lambda$ for any high enough $i < \kappa$.

- $p_{i+1} \restriction (\nu_{i+1} + 1) = p_i \restriction (\nu_{i+1} + 1)$ (in particular, $\nu_{i+1} \notin \text{supp}(p_{i+1})$).
- $p_{i+1} \setminus (\nu_{i+1} + 1)$ extends $p_i \setminus (\nu_{i+1} + 1)$, and satisfies that

$$\{r \in \mathbb{P} \restriction \nu_{i+1} + 1 : r \restriction (p_{i+1} \setminus (\nu_{i+1} + 1)) \in d(\nu_{i+1})\}$$

is a dense subset of $\mathbb{P} \restriction \nu_{i+1} + 1$.

Note that the amount of closure of $\mathbb{P} \setminus \nu_{i+1} + 1$ is larger than $|\mathbb{P} \restriction \nu_{i+1} + 1|$, allowing the construction of such a condition p_{i+1} . Finally, let $C_{i+1} \subseteq C_i$ be a club in κ disjoint from $\text{supp}(p_{i+1})$.

For the limit step, assume that $i < \kappa$ and $\langle p_j, \nu_j, C_j : j < i \rangle$ were constructed. First, set $\nu_i = \sup\{\nu_j : j < i\}$. Let $p_i \in \mathbb{P}$ be a condition such that:

- $p_i \restriction \nu_i = \bigcup_{j < i} p_j \restriction (\nu_j + 1)$.
- $\nu_i \notin \text{supp}(p_i)$.
- $p_i \setminus (\nu_i + 1)$ extends all the conditions $\langle p_j \setminus (\nu_i + 1) : j < i \rangle$, and satisfies that

$$\{r \in \mathbb{P} \restriction \nu_i + 1 : r \restriction (p_i \setminus (\nu_i + 1)) \in d(\nu_i)\}$$

is a dense subset of $\mathbb{P} \setminus \nu_i + 1$.

Note that $\mathbb{P} \setminus \nu_i + 1$ is closed enough to fulfill the third requirement. Also, by our construction, ν_i is a limit point of each club C_j for $j < i$, and thus $\nu_i \notin \text{supp}(p_j)$ for every $j < i$. This ensures that p_i extends p_j for $j < i$. Finally, let $C_i \subseteq \bigcap_{j < i} C_j$ be a club disjoint from $\text{supp}(p_i)$. This concludes the inductive construction, which concludes the proof. $\square$

Corollary 1.4. *Let $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ be as in Lemma 1.3, $p \in \mathbb{P}$ and let $\dot{f}$ be a $\mathbb{P}$ -name forced by p to be a function from κ to the ordinals. Then there exists $q \geq p$, a club $C \subseteq \kappa$ and a function $F \in V$ such that $C \subseteq \text{dom}(F)$ and for every $\alpha \in C$, $|F(\alpha)| \leq |\mathbb{P}_{\alpha+1}|$ (in particular, $|F(\alpha)| < \min(I \setminus \alpha + 1)$) and $q \Vdash \dot{f}(\alpha) \in F(\alpha)$.*

Proof. Apply Lemma 1.3 on the sequence $\langle d(\alpha) : \alpha < \kappa \rangle$, where each $d(\alpha)$ is the dense-open set of conditions deciding the value of $\dot{f}(\alpha)$. Let $q \geq p$ be the obtained condition and $C \subseteq \kappa$ the obtained club, and define, for every $\alpha \in C$,

$$F(\alpha) = \{\gamma \in \text{ON} : \exists r \geq q \restriction \alpha + 1 (r \restriction (q \setminus \alpha + 1)) \Vdash \dot{f}(\alpha) = \check{\gamma}\}.$$

$\square$

The assumptions of Lemma 1.3 suffice to ensure that κ is preserved after forcing with $\mathbb{P}$. But we obtain even more.

Corollary 1.5. *Let $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ be as in 1.3. Then there is no cofinal function $f : \kappa \rightarrow \kappa^+$ in $V[G]$.*

Proof. Let $p \in \mathbb{P}$ and $\dot{f}$ be a $\mathbb{P}$ -name for an increasing function from κ to κ^+ . We wish to prove that $\text{Im}(\dot{f})$ is forced by an extension of p to be bounded in κ^+ . By 1.4 there exists $q \geq p$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$, $q \Vdash \dot{f}(\alpha) \in F(\alpha)$ and $|F(\alpha)| < \min(I \setminus (\alpha+1)) < \kappa$. Let $\gamma = \sup\{\sup(F(\alpha)) : \alpha \in C\} + 1$. Then $\gamma < \kappa^+$ and, since C is unbounded in κ and $\dot{f}$ is increasing, $q \Vdash \sup(\text{Im}(\dot{f})) < \gamma$. $\square$

In particular, Corollary 1.5 implies that $\mathbb{P}$ preserves κ^+ . Assuming further that GCH holds in V , and that each participating forcing $\dot{\mathbb{Q}}_\alpha$ preserves cardinals, we can strengthen the conclusion to full preservation of cardinals by $\mathbb{P}$. Without the GCH assumption, however, this need not hold (see Claim 5.4).

Corollary 1.6. *Assume GCH. Let $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ be as in 1.3. Assume that for every $\alpha \in I$, $\Vdash_{\mathbb{P}_\alpha} \text{"}\mathbb{Q}_\alpha \text{ preserves cardinals"}$. Then $\mathbb{P}$ preserves cardinals.*

Proof. Assume otherwise, and let μ be the least cardinal which is not preserved. It follows that μ is a successor cardinal. We can't have $\mu \geq \kappa^{++}$ since $|\mathbb{P}| = \kappa^+$ by our assumption that $2^\kappa = \kappa^+$. So $\mu = \lambda^+$ for some $\lambda \leq \kappa$, and by Corollary 1.5, we can assume that $\lambda < \kappa$. The forcing $\mathbb{P} \restriction (\lambda+1)$ is more than λ^+ -closed, and, if $\lambda \in I$, the forcing $\mathbb{Q}_\lambda$ preserves λ^+ . So if $\mu = \lambda^+$ is collapsed, this is due to $\mathbb{P} \restriction \lambda$. If λ is not a limit point of I , then by GCH, $|\mathbb{P} \restriction \lambda| \leq \lambda^+$ and thus μ is preserved. So we can assume that λ is a limit point of I , and $\mathbb{P}_\lambda$ collapses $\mu = \lambda^+$. This contradicts Corollary 1.5. $\square$

We conclude this section with the following classical application of Corollary 1.4.

Corollary 1.7. *Assume that κ is regular. For every club $C \subseteq \kappa$ in $V[G]$, there exists a club $D \subseteq \kappa$ in V such that $D \subseteq C$.*

Proof. Let $f \in V[G]$ be the increasing enumeration of C . Apply Corollary 1.4 to find, in V , a function $g: \kappa \rightarrow \kappa$ such that g dominates f on a (ground model) club. Since f is increasing and κ is regular, we can assume that g is increasing and dominates f everywhere. Now define D to be the club of closure points of g . Then $D \subseteq C$, since every $\alpha \in D$ is a limit point of $C \cap \alpha$: indeed, if $i < \alpha$ then $f(i) \leq g(i) < \alpha$ and $i \leq f(i) \in C$. $\square$

2 The splitting forcing

The goal of this section is to prove Theorem 0.2.

Let κ be a measurable in V . Denote by I the set of inaccessible cardinals below κ . Fix throughout this section an ordinal $\tau \leq \kappa^+$. For every $\eta < \kappa^+$, let f_η be the η -th canonical function. Let f_{κ^+} be the successor cardinal function.

Let us define the main forcing notion in this paper. We refer to it as "the splitting forcing".

Definition 2.1 (The Splitting forcing). *Define*

$$\mathbb{P}^\tau = \{f \in \prod_{\alpha \in X} f_\tau(\alpha) : X = \text{dom}(f) \text{ is a nowhere stationary subset of } I\}$$

ordered by inclusion, that is, f extends g if and only if $g \subseteq f$.

Note that in the case $\tau < \kappa$, $\mathbb{P}^\tau$ admits the simpler presentation

$$\mathbb{P}^\tau = \{f : X \rightarrow \tau : X = \text{dom}(f) \text{ is a nowhere stationary subset of } I\}.$$

The forcing notion $\mathbb{P}^\tau$ can be viewed as a nonstationary support product $\prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$, where, for every $\alpha < \kappa$, $\mathbb{Q}_\alpha$ is trivial unless α is inaccessible, and then $\mathbb{Q}_\alpha = \{0_{\mathbb{Q}_\alpha}\} \cup f_\tau(\alpha)$ is the atomic forcing notion that satisfies:

- $0_{\mathbb{Q}_\alpha}$ is the weakest condition of $\mathbb{Q}_\alpha$.
- the ordinals in $f_\tau(\alpha)$ form a maximal antichain in $\mathbb{Q}_\alpha$.

In particular, a generic for $\mathbb{Q}_\alpha$ is basically a choice of an ordinal in $f_\tau(\alpha)$.

Remark 2.2. *There are two main reasons for the limitation $\tau \leq \kappa^+$ in this section.*

1. *As a nonstationary-support product, $\mathbb{P}^\tau$ can be analyzed with the Fusion Lemma 1.3. To use it, however, we must ensure that $\mathbb{Q}_\alpha \in V_\kappa$ for all α . If $2^\kappa = \kappa^+$, then $\tau = \kappa^{++}$ cannot be represented by a function f_τ whose values lie below κ , and the forcings $\mathbb{Q}_\alpha$ turn out to be quite large. On the other hand, if GCH fails at κ and U is a normal measure that computes κ^{++} correctly in its ultrapower, then forcing with $\mathbb{P}^\tau$ may collapse κ^{++} (see Claim 5.4).*
2. *The first reason doesn't rule out the possibility that τ is any ordinal below the least inaccessible above κ in $\text{Ult}(V, U)$, where U is the normal measure we would like to lift.⁷ Relaxing the restriction $\tau < \kappa^+$ makes the classification of normal measures more complicated than the one exhibited in Theorem 0.2. We study this more general framework in section 4, and produce the classification of normal measures in Theorem 4.4.*

Lemma 2.3. *Assume that $G \subseteq \mathbb{P}^\tau$ is generic over V . For every $\eta < \tau$, define–*

$$S_\eta = \{\alpha \in I : (\cup G)(\alpha) = f_\eta(\alpha)\}$$

Then $\vec{S} = \langle S_\eta : \eta < \tau \rangle$ is a sequence of stationary subsets of I . If $\tau < \kappa$, $\vec{S}$ is a partition of I and $V[G] = V[\vec{S}]$.

⁷Working with such τ , we can ensure that $|\mathbb{Q}_\alpha|$ is below the amount of closure of the forcing $\mathbb{P}^\tau \restriction \alpha + 1$, which is one of the assumptions of the Fusion Lemma.

Proof. Fix $\eta < \tau$, and let $C \in V[G]$ be a club in κ . By Corollary 1.7 there exists in V a club $D \subseteq \kappa$ such that $D \subseteq C$. Given a condition $p \in \mathbb{P}^\tau$, we can pick $\alpha \in D \setminus \text{supp}(p)$ due to the nonstationary support, and extend p to $p \cup \{(\alpha, f_\eta(\alpha))\}$. Thus, by density, there exists $q \in G$ and $\alpha \in D$ such that $q(\alpha) = f_\eta(\alpha)$. Thus, in $V[G]$, $\alpha \in C \cap S_\eta$.

If $\tau < \kappa$ then each $f_\eta(\alpha)$ simply equals η , and $S_\eta, S_{\eta'}$ for $\eta \neq \eta'$ are disjoint. Also, G could be read from $\vec{S}$, since for every condition p , $p \in G$ if and only if

$$\forall \alpha \in \text{dom}(p) \ (p(\alpha) = \eta \iff \alpha \in S_\eta).$$

□

Next, we show that every normal measure $U \in V$ on κ generates, together with one of the stationary sets S_η , a normal measure on κ in $V[G]$.

Lemma 2.4. *Let $U \in V$ be a normal measure on κ . Let $G \subseteq \mathbb{P}^\tau$ be generic over V . Then for every $\eta < \tau$, $U \cup \{S_\eta\}$ generates a normal measure U_η^* on κ in $V[G]$. Furthermore, the U_η^* -s have the same ultrapower over $V[G]$.*

Proof. Since τ is fixed, we denote $\mathbb{P} = \mathbb{P}^\tau$ throughout this proof. Fix $\eta < \tau$. The forcing $j_U(\mathbb{P})$ is, by elementarity, a nonstationary support product $\prod_{\alpha \in j_U(I)}^{NS} \mathbb{Q}_\alpha^{M_U}$, where, for each $\alpha \in j_U(I) \cap \kappa = I$, $\mathbb{Q}_\alpha^{M_U} = \mathbb{Q}_\alpha$, and for each $\alpha \in j_U(I) \setminus \kappa$, $\mathbb{Q}_\alpha^{M_U}$ is an atomic forcing that chooses an ordinal below $j_U(f_\tau)(\alpha)$. In particular, $\kappa \in j_U(I)$ and $\mathbb{Q}_\kappa^{M_U}$ is an atomic forcing that chooses an ordinal below τ . Let

$$H_\eta = \{q \in j_U(\mathbb{P}) : \exists p \in G \ (q \subseteq j_U(p) \cup \{\langle \kappa, \eta \rangle\})\}.$$

Note that for every $p \in \mathbb{P}$, $\kappa \notin \text{supp}(j_U(p))$. Therefore $j_U(p) \cup \{\langle \kappa, \eta \rangle\}$ is a function and its support is a nowhere stationary subset of $j_U(\kappa)$, so $H_\eta \subseteq j_U(\mathbb{P})$. Furthermore, the elements of H_η are mutually compatible and clearly $j_U[G] \subseteq H_\eta$. We argue that $H_\eta \subseteq j_U(\mathbb{P})$ is generic over M_U .

Fix $E \subseteq j_U(\mathbb{P})$ dense open in M_U . Let $\langle e(\alpha) : \alpha < \kappa \rangle$ be a sequence of dense open subsets of $\mathbb{P}$ such that $E = [\alpha \mapsto e(\alpha)]_U$. Recall that $\mathbb{P}$ can be viewed as a nonstationary support product. Apply the Fusion Lemma 1.3 to find a condition $q \in G$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$,

$$\{r \in \mathbb{P}_{\alpha+1} : r^\frown(q \setminus \alpha + 1) \in e(\alpha)\}$$

is dense in $\mathbb{P}_{\alpha+1}$. This means that, in M_U , the set–

$$\{r \in j_U(\mathbb{P})_{\kappa+1} : r^\frown(j_U(q) \setminus \alpha + 1) \in E\}$$

is dense. Since G is generic for $j_U(\mathbb{P})_\kappa = \mathbb{P}_\kappa$ (over M_U), and η is generic for $\mathbb{Q}_\kappa^{M_U}$ (over $M_U[G]$), there exists $p \in G$ such that $p^\frown \langle \langle \kappa, \eta \rangle \rangle^\frown (j_U(q) \setminus \kappa + 1) \in E$. By increasing p if necessary inside G , we can assume that p extends q . Since E is open, we deduce that $j_U(p) \cup \{\langle \kappa, \eta \rangle\} \in E \cap H_\eta$, as desired.

Since $j_U[G] \subseteq H_\eta$, we can now use Silver's criterion to lift the embedding j_U to an elementary embedding from $V[G]$ to $M_U[H_\eta]$. By standard arguments (see [8]), the lifted embedding is an ultrapower embedding with respect to a normal measure U_η^* . Indeed, U_η^* is the normal measure derived from the lifted embedding using κ as a seed. In other words, for $X \in V[G]$, we have $X \in U_\eta^*$ if and only if there exists a condition in H_η that forces that $\kappa \in j_U(\dot{X})$ (where $\dot{X}$ is a $\mathbb{P}$ -name for X). Equivalently, there exists a condition $p \in G$ such that

$$j_U(p) \cup \{\langle \kappa, \eta \rangle\} \Vdash \kappa \in j_U(\dot{X}).$$

This implies that $\{\alpha < \kappa : p \cup \{\langle \alpha, f_\eta(\alpha) \rangle\} \in U\}$, and

$$\{\alpha < \kappa : p \cup \{\langle \alpha, f_\eta(\alpha) \rangle\} \Vdash \check{\alpha} \in \dot{X}\} \cap \{\alpha < \kappa : (\cup G)(\alpha) = f_\eta(\alpha)\} \subseteq X$$

so $U \cup \{S_i\}$ generates U_i^* .

Note that for every $\eta, \zeta < \tau$, H_η, H_ζ are mutually definable: one is obtained from the other by changing the generic value at κ .⁸ This implies that all the measures U_η ($\eta < \tau$) have the same ultrapower. $\square$

We now proceed to prove that every normal measure of $V[G]$ has the form U_η^* for some $\eta < \tau$ and normal measure $U \in V$. The main tool that assists bypassing inner-model-theoretic arguments here, is Hamkins' Gap forcing theorem.

Theorem 2.1 (Hamkins' Gap Forcing theorem [13]). *Assume that $\mathbb{P}$ is a forcing notion which has a gap at some cardinal δ : that is, $\mathbb{P}$ can be factored to the form $\mathbb{P} = \mathbb{P}_0 * \mathbb{P}_1$, where $\mathbb{P}_0$ is nontrivial, $|\mathbb{P}_0| < \delta$ and $\Vdash_{\mathbb{P}_0} \mathbb{P}_1$ is $(\delta + 1)$ -strategically closed." Let $G \subseteq \mathbb{P}$ be generic over V , and assume that $j : V[G] \rightarrow M^*$ is an elementary embedding with critical point $\kappa > \delta$, such that $M^* \subseteq V[G]$ and M^* is closed under δ -sequences of its elements that belong to $V[G]$. Then:*

1. M^* has the form $M[H]$ where $M \subseteq V$ and $H = j(G)$ is $j(\mathbb{P})$ -generic over M . Furthermore, $M = V \cap M[H]$.
2. If j is definable in $V[G]$ from parameters, then $j \restriction V : V \rightarrow M$ is definable in V from parameters.

We remark that M^* having the form $M[H]$ already follows from Laver's ground model definability theorem. However, the inclusion $M \subseteq V$ is nontrivial and need not hold for forcings without a gap (see, for example, [10, Subsection 5.2]).

Remark 2.5. *Nonstationary support iterated forcings usually satisfy the assumptions of the Gap forcing theorem. Our splitting forcing $\mathbb{P}^\tau$, however, is a*

⁸The fact that we used a product rather than an iteration matters here: if we forced with an iteration, and $q' \in H_\eta$ is obtained from q by replacing $\langle \kappa, \eta \rangle$ (assuming that $q \restriction \kappa$ forces that $\langle \kappa, \eta \rangle$ belongs to q) with $\langle \kappa, \zeta \rangle$, the fact that $q \subseteq j_U(p) \cup \{\langle \kappa, \eta \rangle\}$ for some $p \in G$ does not necessarily imply that $q' \subseteq j_U(p') \cup \{\langle \kappa, \zeta \rangle\}$ for some $p' \in G$. Using products ensures that $q \restriction \kappa + 1$ doesn't depend on $q(\kappa)$, and thus q' is compatible with H_ζ .

nonstationary-support product rather than an iteration. It is unclear whether the conclusions of the Gap Forcing Theorem apply directly to $\mathbb{P}^\tau$. To address this, we modify $\mathbb{P}^\tau$ by considering $\mathbb{R}^\tau = \text{Cohen}(\omega) * \mathbb{P}^\tau$, which has a gap at $\delta = \omega_1$. An alternative approach would be to define $\mathbb{P}^\tau$ as a nonstationary-support iteration instead of a product, but this comes at the cost that the ultrapowers $\text{Ult}(V[G], U_\eta^*)$ for $\eta < \tau$ are not necessarily the same (this would still provide a proof for the original Friedman-Magidor theorem 0.1).

With this in mind, we now have all the ingredients to conclude the proof of Theorem 0.2.

Proof of Theorem 0.2. Let $\mathbb{R}^\eta = \text{Cohen}(\omega) * \mathbb{P}^\tau$. Let $G = g * G' \subseteq \mathbb{R}^\tau$ be generic over V , where g is Cohen-generic over V , and G' is $\mathbb{P}^\tau$ -generic over $V[g]$. By the Lévy–Solovay theorem [15], every normal measure on κ in V generates a normal measure on κ in $V[g]$, and conversely, every normal measure on κ in $V[g]$ arises in this way. Given $U \in V$ a normal measure, denote by $U' \in V[g]$ the normal measure it generates. As above, given $\eta < \tau$, let $S_\eta = \{\alpha < \kappa : (\cup G')(\alpha) = f_\eta(\alpha)\}$. By Lemma 2.4, for every $\eta < \tau$, $U' \cup \{S_\eta\}$ generates a measure $U'_\eta \in V[g * G']$, and all the measures $\langle U'_\eta : \eta < \tau \rangle$ have the same ultrapower. We can slightly abuse the notation and denote U_η^* instead of U'_η . Each such U_η^* is generated from $U \cup \{S_\eta\}$ in $V[g * G']$, since every set in U' contains a set from U . This concludes the proof of clause 1 of Theorem 0.2.

We proceed and prove clause 2. Let $W \in V[G]$ be a normal measure on κ , where, as before, G is viewed as a generic $g * G' \subseteq \text{Cohen}(\omega) * \mathbb{P}^\tau$. By Hamkins' Gap forcing theorem, $U = W \cap V \in V$, since $U = \{X \subseteq \kappa : \kappa \in (j_W \upharpoonright V)(X)\}$. U is normal because W is normal in $V[G]$. Denote $\eta = j_W(\cup G')(\kappa)$. Using again the normality of W , $\eta < j_W(f_\tau)(\kappa) = \tau$ and $S_\eta \in W$. So $U \cup \{S_\eta\} \subseteq W$, and, by the previous paragraph, $W = U_\eta^*$.

Finally, if GCH holds in V , then it continues to hold in $V[g]$, and over this model $\mathbb{P}^\tau$ preserves cardinals by Corollary 1.6. $\square$

We finish this section by proving that the splitting forcing $\mathbb{P}^\tau$ preserves the Mitchell order. As a corollary, $\mathbb{R}^\tau$ preserves the Mitchell order as well.

Lemma 2.6. *Assume that $U_0 \triangleleft U_1$ are normal measures on κ in V . Fix $\eta, \zeta < \tau$ and $G \subseteq \mathbb{P}^\tau$ generic over V . Then $(U_0)_\eta^* \triangleleft (U_1)_\zeta^*$.*

Proof. Since $U_0 \triangleleft U_1$, we can lift U_0 to $\left((U_0)_\eta^*\right)^{M_{U_1}[G]}$ by forcing with $\mathbb{P}^\tau$ over M_{U_1} . It suffices to prove that $\left((U_0)_\eta^*\right)^{M_{U_1}[G]} = (U_0)_\zeta^*$.

Both ultrafilters $\left((U_0)_\eta^*\right)^{M_{U_1}[G]}$, $(U_0)_\zeta^*$ are generated by $U_0 \cup \{S_\eta\}$, but the former ultrafilter belongs to $V[G]$ and the latter belongs to $M_{U_1}[G] \subseteq V[G]$. So we just need to verify that $V[G]$ and $M_{U_1}[G]$ share the same subsets of κ . This is another classical application of the Fusion Lemma 1.3.

Let $\dot{X} \in V$ be a $\mathbb{P}^\tau$ -name for a subset of κ . For every $\alpha < \kappa$, let

$$e(\alpha) = \{r \in \mathbb{P}^\tau : r \upharpoonright \alpha + 1 \Vdash \exists X_\alpha \subseteq \alpha \left(r \setminus (\alpha + 1) \Vdash \dot{X} \cap \alpha = X_\alpha \right)\}.$$

Each $e(\alpha)$ is a dense open subset of $\mathbb{P}^\tau$, since $\mathbb{P}^\tau \setminus \alpha + 1$ is more than α -closed. By the Fusion Lemma 1.3, there exists $p \in G$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$,

$$p \restriction \alpha + 1 \Vdash \exists X_\alpha \subseteq \alpha \left(p \restriction (\alpha + 1) \Vdash \dot{X} \cap \alpha = X_\alpha \right).$$

For each $\alpha \in C$, let $\dot{X}_\alpha$ be a canonical $\mathbb{P}_{\alpha+1}$ -name for X_α as a subset of α . Note that $\langle X_\alpha : \alpha \in C \rangle \in M_{U_1}$ and also $p \in M_{U_1}$. Since $p \in G$, we can compute $(\dot{X})_G$ in $M_{U_1}[G]$ from G and $\langle \dot{X}_\alpha : \alpha \in C \rangle$. $\square$

3 Lifting iterates of a normal measure

Our next goal is to prove Theorem 0.4, which provides a characterization of all possible lifts of U^n for a normal measure U on κ and $n < \omega$.

We maintain the same notation as in the previous section, including the measurable cardinal κ and the forcing notion $\mathbb{P}^\tau$, which we denote simply by $\mathbb{P}$ throughout this section.⁹

In this section, we restrict attention to the case $\tau < \kappa$ only, due to technical difficulties that arise when applying the same proof technique in the cases $\tau \in \{\kappa, \kappa^+\}$.

We denote by κ_i for $i < \omega$ the images of κ under the iterated ultrapowers with U ; that is, for each $i < \omega$, $\kappa_i = j_{U^i}(\kappa)$.

Lemma 3.1. *Fix a normal measure U on κ , $n < \omega$, and a sequence $\langle \eta_0, \dots, \eta_{n-1} \rangle \in {}^\tau \kappa$. Assume that for every $\vec{\nu} = \langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n$, $e(\vec{\nu}) = e(\nu_0, \dots, \nu_{n-1}) \subseteq \mathbb{P}$ is a dense open set. Then for every $p \in \mathbb{P}$ there exists $r \geq p$ such that*

$$\{\vec{\nu} \in [\kappa]^n : r \cup \{\langle \nu_0, \eta_0 \rangle, \langle \nu_1, \eta_1 \rangle, \dots, \langle \nu_{n-1}, \eta_{n-1} \rangle\} \in e(\vec{\nu})\} \in U^n.$$
¹⁰

Proof. We begin with the case $n = 1$. Let p be a condition. By the Fusion Lemma 1.3, there exists a condition $q \geq p$ and a club $C \subseteq \kappa$ such that for every $\nu \in C$,

$$\{r \in \mathbb{P}_{\nu+1} : r \cup (q \restriction \nu + 1) \in e(\nu)\}$$

is a dense subset of $\mathbb{P}_{\nu+1}$ above $q \restriction \nu + 1$. We can assume that $C \cap \text{supp}(q) = \emptyset$ by shrinking C . Then for every $\nu \in C$, there exists a condition $r(\nu) \in \mathbb{P}_\nu$ extending $q \restriction \nu$ such that

$$r(\nu) \cup \{\langle \nu, \eta_0 \rangle\} \cup (q \restriction \nu + 1) \in e(\nu).$$

Since $j_U(\mathbb{P}) \restriction \kappa = \mathbb{P}$ and $j_U(q) \restriction \kappa = q$, we deduce that $r = [\nu \mapsto r(\nu)]_U$ is a condition in $\mathbb{P}$ extending q . Furthermore, there exists a set $A \in U$ such that for every $\nu \in A$, $r \restriction \nu = r(\nu)$. Then for every $\nu \in A \cap C \in U$,

$$r \restriction \nu \cup \{\langle \nu, \eta_0 \rangle\} \cup (r \restriction \nu + 1) \in e(\nu).$$

⁹In this section, there is no need to force with $\mathbb{R}^\tau = \text{Cohen}(\omega) * \mathbb{P}^\tau$, as the main result is obtained by forcing over $L[U]$, and standard arguments eliminate the need for Hamkins' Gap Forcing Theorem.

¹⁰Note that there exists a club C disjoint from $\text{supp}(r)$, and $[C]^n \in U^n$. So we can assume that $r \cup \{\langle \nu_0, \eta_0 \rangle, \dots, \langle \nu_{n-1}, \eta_{n-1} \rangle\}$ is a function for a set of $\vec{\nu}$ -s in U^n .

Now proceed by induction. Fix a sequence $\langle \eta_0, \dots, \eta_{n-1}, \eta_n \rangle \in \tau^{n+1}$. Assume that for every $\langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n$, and for every $\nu \in \kappa \setminus \nu_{n-1} + 1$,

$$e(\nu_0, \dots, \nu_{n-1}, \nu) \subseteq \mathbb{P}$$

is a dense open set.

Fix $\vec{\nu} = \langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n$. Define set $e(\nu_0, \dots, \nu_{n-1})$ consisting of conditions r for which there exists a set $A \in U$, such that for every $\nu \in A \setminus \nu_{n-1} + 1$,

$$r \cup \{\langle \nu, \eta_n \rangle\} \in e(\nu_0, \dots, \nu_{n-1}, \nu).$$

Applying the case $n = 1$, with η_n in the role of η_0 , shows that each set $e(\nu_0, \dots, \nu_{n-1})$ is a dense open subset of $\mathbb{P}$.

Fix $p \in \mathbb{P}$. Apply the induction hypothesis on the sequence

$$\langle e(\nu_0, \dots, \nu_{n-1}) : \langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n \rangle.$$

There are $r \geq p$ and $X \in U^n$ such that for every increasing sequence $\vec{\nu} = \langle \nu_0, \dots, \nu_{n-1} \rangle \in X$,

$$r \cup \{\langle \nu_0, \eta_0 \rangle, \dots, \langle \nu_{n-1}, \eta_{n-1} \rangle\} \in e(\vec{\nu}).$$

In particular, for every such sequence $\langle \nu_0, \dots, \nu_{n-1} \rangle$ there exists a set $A(\nu_0, \dots, \nu_{n-1}) \in U$ such that for every $\nu_n \in A(\nu_0, \dots, \nu_{n-1}) \setminus \nu_{n-1} + 1$,

$$r \cup \{\langle \nu_0, \eta_0 \rangle, \dots, \langle \nu_{n-1}, \eta_{n-1} \rangle, \langle \nu, \eta_n \rangle\} \in e(\nu_0, \dots, \nu_{n-1}, \nu_n).$$

Let

$$Y = \{\langle \nu_0, \dots, \nu_{n-1}, \nu \rangle \in [\kappa]^{n+1} : \langle \nu_0, \dots, \nu_{n-1} \rangle \in X \text{ and } \nu \in A(\nu_0, \dots, \nu_{n-1})\}$$

It's not hard to see that $Y \in U^{n+1}$. Shrink Y by intersecting it with $[C]^{n+1}$, where C is some club disjoint from $\text{supp}(r)$. It follows that the condition r has the desired property, witnessed by the shrunk version of Y which belongs to U^{n+1} . $\square$

Corollary 3.2. *Let U be a normal measure on κ . Fix $n < \omega$ and $\tau < \kappa$. Let $G \subseteq \mathbb{P}^\tau$ be generic over V . Then for every sequence $\vec{\eta} = \langle \eta_0, \dots, \eta_{n-1} \rangle \in \tau^n$, the set $U^n \cup \{S_{\vec{\eta}}\}$ generates a κ -complete ultrafilter in $V[G]$, where*

$$S_{\vec{\eta}} = \{\langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n : \forall i < n, (\cup G)(\nu_i) = \eta_i\}.$$

Proof. We construct a $j_{U^n}(\mathbb{P})$ -generic set over M_{U^n} . Let

$$H_{\vec{\eta}} = \{q \in j_{U^n}(\mathbb{P}) : \exists p \in G (q \leq j_{U^n}(p) \cup \{\langle \kappa, \eta_0 \rangle, \dots, \langle \kappa_{n-1}, \eta_{n-1} \rangle\})\}.$$

We argue that $H_{\vec{\eta}} \subseteq j_{U^n}(\mathbb{P})$ is generic over M_{U^n} .

First, note that for every $m \leq n$ and $p \in \mathbb{P}$, $\kappa_m \notin j_{U^n}(p)$ (since κ_m belongs to $j_{U^n}(C)$ where $C \subseteq \kappa$ is any club disjoint from the support of p). Thus, $H_{\vec{\eta}}$ consists of legitimate conditions in $j_{U^n}(\mathbb{P})$. Clearly any pair of conditions in $H_{\vec{\eta}}$

are compatible. Thus, we only need to argue that $H_{\vec{\eta}}$ is $j_{U^n}(\mathbb{P})$ -generic over V . This is an immediate corollary of Lemma 3.1: given any dense open set $E \subseteq j_{U^n}(\mathbb{P})$ in M_{U^n} , let $\langle e(\nu_0, \dots, \nu_{n-1}) : \langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n \rangle$ be a sequence of dense open subsets of $\mathbb{P}$ such that

$$E = j_{U^n}(\langle \nu_0, \dots, \nu_{n-1} \rangle \mapsto e(\nu_0, \dots, \nu_{n-1}))(\kappa_0, \dots, \kappa_{n-1}).$$

Pick a condition $r \in G$ such that

$$\{\vec{\nu} \in [\kappa]^n : r \cup \{\langle \nu_0, \eta_0 \rangle, \dots, \langle \nu_{n-1}, \eta_{n-1} \rangle\} \in e(\nu_0, \dots, \nu_{n-1})\} \in U^n.$$

Then $j_{U^n}(r) \cup \{\langle \kappa, \eta_0 \rangle, \dots, \langle \kappa_{n-1}, \eta_{n-1} \rangle\} \in E \cap H_{\vec{\eta}}$, as desired. $\square$

As a corollary, we obtain a complete characterization of all possible lifts of U^n in $V[G]$.

Corollary 3.3. *Let U be a normal measure on κ . Fix $n < \omega$ and $\tau < \kappa$. Let $G \subseteq \mathbb{P} = \mathbb{P}^\tau$ be generic over V . For every σ -complete ultrafilter $W \in V[G]$ such that $U^n \subseteq W$, there exists $\vec{\eta} = \langle \eta_0, \dots, \eta_{n-1} \rangle \in \tau^n$ such that W is the measure generated by $U^n \cup \{S_{\vec{\eta}}\}$.*

Proof. Let $j_W : V[G] \rightarrow M[H] = M[j_W(G)]$ be the ultrapower embedding associated with W .

Since $U^n \subseteq W$, W concentrates on the set $[I]^n$ consisting of increasing sequences of inaccessibles (in V) of length n . Thus, $[Id]_W = \{\lambda_0, \dots, \lambda_{n-1}\}$ for some inaccessibles (of M) $\lambda_0 < \dots < \lambda_{n-1}$.

It follows that the generic $H = j_W(G)$ assigns to each λ_i (for $0 \leq i \leq n-1$) some $\eta_i = (\cup H)(\lambda_i) < j_W(\tau) = \tau$. Thus, letting $\vec{\eta} = \langle \eta_0, \dots, \eta_{n-1} \rangle \in \tau^n$, we deduce that $S_{\vec{\eta}} \in W$, and W is the measure generated by $U^n \cup \{S_{\vec{\eta}}\}$. $\square$

All the results provided so far have not required the assumption that $V = L[U]$. However, to obtain a full characterization of all σ -complete ultrafilters, the assumption $V = L[U]$ is useful, since it implies that every σ -complete ultrafilter is Rudin–Keisler equivalent to one extending U^n for some $n < \omega$. The proof of Theorem 0.4, given below, explains this in detail.

Proof of Theorem 0.4. Assume $V = L[U]$ is the core model. Let $G \subseteq \mathbb{P}^\tau$ be generic over $L[U]$.

We first show that every σ -complete ultrafilter $W \in L[U][G]$ on an ordinal χ satisfies $W \cap L[U] \in L[U]$. Indeed, the restriction $j_W \upharpoonright L[U]$ is an iterated ultrapower j_{U^m} of U over $L[U]$ for some $m < \omega$. Consequently,

$$W \cap L[U] = \{X \subseteq \chi : [Id]_W \in j_{U^m}(X)\} \in L[U].$$

For clause 1 of Theorem 0.4, the argument proceeds exactly as in the proof of Theorem 0.2: given a normal measure $W \in L[U][G]$ on κ , first observe that $W \cap L[U]$ inherits normality from W , so $U = W \cap L[U]$; then set $\eta = j_W(\cup G)(\kappa)$, note that $S_\eta \in W$, and conclude that W is the normal measure U_η^* generated by $U \cup \{S_\eta\}$.

Let us proceed and prove clauses 2 and 3. Let $W \in V[G]$ be an arbitrary σ -complete ultrafilter in $V[G]$, concentrating on an ordinal χ . By the previous paragraphs, $W \cap L[U]$ is a σ -complete ultrafilter in $L[U]$, and as such, it is Rudin-Keisler equivalent to U^n for some $n < \omega$. Fix an injection $h: \chi \rightarrow [\kappa]^n$ witnessing this (namely, for every set $X \subseteq \chi$, $X \in W \cap L[U]$ if and only if $h[X] \in U^n$). By replacing W with $h_*(W) = \{X \subseteq [\kappa]^n : h^{-1}[X] \in W\}$, we can assume from now on W concentrates on $[\kappa]^n$ and $W \supseteq U^n$. This is harmless since our goal is characterizing j_W and $\text{Ult}(L[U][G], W)$, which are invariant under Rudin-Keisler equivalence. By Corollary 3.3, there exists $\vec{\eta} = \langle \eta_0, \dots, \eta_{n-1} \rangle \in \tau^n$ such that W is the measure generated by $U^n \cup \{S_{\vec{\eta}}\}$.

We proceed and prove that $\text{Ult}(L[U][G], W)$ is a finite iterated ultrapower of $L[U][G]$ via normal measures.

Let $W^0 \in V[G]$ be the normal measure on κ generated from $U \cup \{S_{\eta_0}\}$ (in the notations of Lemma 2.4). Denote $N^0 = \text{Ult}(V[G], W^0)$, and note that $N^0 = M_U[H_{\eta_0}]$, where $H_{\eta_0} = j_{W^0}(G)$ is generated from $(\cup j_U[G]) \cup \{\langle \kappa, \eta_0 \rangle\}$.

Over $N^0 = M_U[H_{\eta_0}]$, let W^1 be the normal measure on κ_1 generated from $j_U(U) \cup \{S_{\eta_1}^1\}$, where $S_{\eta_1}^1 = \{\alpha < \kappa_1 : (\cup H_{\eta_0})(\alpha) = \eta_1\}$. Set $N^1 = \text{Ult}(N^0, W^1)$. We argue that N^1 has the form $M_{U^2}[H_{\langle \eta_0, \eta_1 \rangle}]$ where $H_{\langle \eta_0, \eta_1 \rangle} = j_{W^1}(H_{\eta_1})$ is the $j_{U^2}(\mathbb{P})$ -generic generated by $\{(\cup j_{U^2}[G]) \cup \{\langle \kappa, \eta_0 \rangle, \langle \kappa, \eta_1 \rangle\}\}$.

Indeed, inside $V[G]$, the ground of $\text{Ult}(V[G], W)$ for every normal measure $W \in V[G]$ is definable as $\text{Ult}(V, W \cap V)$ (recall that V itself is definable in $V[G]$ by Laver's ground model definability theorem). Furthermore, $j_W(G)$ is the generic set generated by $(\cup j_{W \cap V}[G]) \cup \{\langle \kappa, \eta \rangle\}$ where η is the unique ordinal below τ such that $S_\eta \in W$. Having all of this in mind, we can now use elementarity to deduce how $N^1 = \text{Ult}(N^0, W^1)$ looks inside $N^0 = M_U[H_{\eta_0}]$: the ground of $\text{Ult}(N^0, W^1)$ is

$$\text{Ult}(M_U, W^1 \cap M_U) = \text{Ult}(M_U, j_U(U)) = M_{U^2}$$

and the corresponding generic set $j_{W^1}(H_{\eta_0})$ is the one generated by

$$j_{W^1 \cap M_U}[H_{\eta_0}] \cup (\{\langle \kappa_1, \eta_1 \rangle\}).$$

Since $W^1 \cap M_U = j_U(U)$ and H_{η_0} itself is generated by $j_U[G] \cup \{\langle \kappa, \eta_0 \rangle\}$, we deduce that $j_{W^1}(H_{\eta_0})$ is the generic generated by

$$j_{U^2}[G] \cup (\{\langle \kappa, \eta_0 \rangle, \langle \kappa_1, \eta_1 \rangle\})$$

which is exactly the generic $H_{\langle \eta_0, \eta_1 \rangle}$ defined above.

Continuing this inductively, we define models N^i , normal measures $W^i \in N^i$, ordinals $\eta_i < \tau$ and generics $H_{\langle \eta_0, \dots, \eta_i \rangle} \subseteq j_{U^i}(\mathbb{P})$, such that for every $i \leq n-1$,

1. $N^i = \text{Ult}(N^i, W^i) \simeq M_{U^{i+1}}[H_{\langle \eta_0, \dots, \eta_i \rangle}]$.
2. W^{i+1} is generated inside N^i from $j_{U^{i+1}}(U) \cup \{S_{\eta_{i+1}}^{i+1}\}$ where $S_{\eta_{i+1}}^{i+1} = \{\alpha < \kappa_{i+1} : (\cup H_{\langle \eta_0, \dots, \eta_i \rangle})(\alpha) = \eta_{i+1}\}$.
3. $H_{\langle \eta_0, \dots, \eta_{i+1} \rangle} \subseteq j_{U^{i+1}}(\mathbb{P})$ is generated from $(\cup H_{\langle \eta_0, \dots, \eta_i \rangle}) \cup \{\langle \kappa_{i+1}, \eta_{i+1} \rangle\}$. Equivalently, it's generated from $j_{U^{i+1}}[G] \cup \{\langle \kappa, \eta_0 \rangle, \dots, \langle \kappa_{i+1}, \eta_{i+1} \rangle\}$.

Note that $H_{\langle \eta_0, \dots, \eta_{n-1} \rangle}$ is the same generic from the proof of Corollary 3.2, corresponding to the ultrapower embedding associated with the measure generated by $U^n \cup \{S_{\langle \eta_0, \dots, \eta_{n-1} \rangle}\}$, which we have already identified as W . It follows that $N^{n-1} = M_{U^n}[H_{\langle \eta_0, \dots, \eta_{n-1} \rangle}]$ is exactly $\text{Ult}(V[G], W)$. In other words, the ultrapower with W is the result of the successive ultrapowers taken with the normal measures $W^0, W^1, \dots, W^{n-1}$ described above. This proves Clause 2 of Theorem 0.4.

Finally, note that for every $i \leq n-1$, $N^i \simeq \text{Ult}(V[G], (U_0^*)^{i+1})$ (where U_0^* is as in Lemma 2.4 for $\eta = 0$). This follows by induction, using the fact that any pair of normal measures in $V[G]$ have the same ultrapower. Indeed, for $i = 0$, $N^0 \simeq \text{Ult}(V[G], W^0) \simeq \text{Ult}(V[G], U_0^*)$. Assuming the statement for i , note that $N^{i+1} \simeq \text{Ult}(N^i, W^i)$. But by the induction hypothesis, $N^i \simeq \text{Ult}(V[G], (U_0^*)^{i+1})$, and $j_{(U_0^*)^{i+1}}(U_0^*)$ is a normal measure on κ_i in N^i . It follows that $N^{i+1} \simeq \text{Ult}(N^i, W^i) \simeq \text{Ult}(N^i, j_{(U_0^*)^{i+1}}(U_0^*)) = \text{Ult}(V[G], (U_0^*)^{i+1})$.

In particular, the above inductive argument shows that N^{n-1} is isomorphic to $\text{Ult}(V[G], (U_0^*)^n)$. This proves Clause 3 of Theorem 0.4. $\square$

4 Friedman–Magidor for extenders

In the previous sections, our goal was to classify lifts of normal measures (and their iterations) in forcing extensions via the splitting forcing $\mathbb{P}^\tau$. We now turn to lifts of extenders. Throughout this section, we say that two extenders E and E' are equivalent if $j_E = j_{E'}$.

For the remainder of the section, fix a function $g: \kappa \rightarrow \kappa$ such that, for every $\alpha \leq \kappa$, $g(\alpha)$ lies below the least inaccessible cardinal above α . Also fix the following more general version of the splitting forcing:

$$\mathbb{P} = \{f \in \prod_{\alpha \in X} g(\alpha) : X = \text{dom}(f) \text{ is nowhere stationary subset of } I\}.$$

As usual, we order $\mathbb{P}$ by inclusion, and define $\mathbb{R} = \text{Cohen}(\omega) * \mathbb{P}$.

Our forcing $\mathbb{P}$ can be viewed as a nonstationary support product $\prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$, where $\mathbb{Q}_\alpha = \{0_{\mathbb{Q}_\alpha}\} \cup g(\alpha)$ is an atomic forcing, in which $g(\alpha) \subseteq \mathbb{Q}_\alpha$ is a maximal antichain. As in Sections 1, 2, GCH ensures that $\mathbb{P}$ and $\mathbb{R}$ preserve cardinals.

The function g determines the number of (nonequivalent) lifts of a given ground model extender to the generic extension. The requirement that each $g(\alpha)$ be below the least inaccessible above α ensures that $\mathbb{P}$ satisfies the assumptions of the Fusion Lemma 1.3. Our method provides control over all lifts of extenders E such that $j_E(g)(\kappa)$ bounds all the generators of E (we conjecture that those methods can be generalized to a wider class of extenders, see Remark 4.1).

Our main goal in this section will be the following theorem:

Theorem 4.1. *Assume GCH, and let $g: \kappa \rightarrow \kappa, \mathbb{P}, \mathbb{R}$ be as above. Let $G \subseteq \mathbb{R}$ be generic over V . Then the following properties hold:*

1. *for every (κ, λ) -extender E in V whose generators are all below $j_E(g)(\kappa)$ and for each $\eta < j_E(g)(\kappa)$, there exists a (κ, λ) -extender $E_\eta^* \in V[G]$ such*

that $j_{E_\eta^*} \restriction V = j_E$. Furthermore, all the extenders $\langle E_\eta^*: \eta < j_E(\tau)(\kappa) \rangle$ have the same ultrapower, and the extenders $\langle E_\eta: \eta < j_E(g)(\kappa) \rangle$ are nonequivalent.

2. Every (κ, λ) -extender $E^* \in V[G]$ whose generators all lie below $j_{E^*}(g)(\kappa)$ and with $\lambda \geq j_{E^*}(g)(\kappa)$ is, up to equivalence, of the form E_η^* , for some (κ, λ) -extender $E \in V$ with generators below $j_E(g)(\kappa)$ and some $\eta < j_E(\tau)(\kappa)$.

We remark that the requirement $\lambda \geq j_{E^*}(g)(\kappa)$ in Clause 2 above is necessary. Indeed, Theorem 4.4 shows that even in the case $\lambda = \kappa + 1$ —that is, when E^* is a normal measure on κ in $V[G]$ —the embedding $j_{E^*} \restriction V$ may have a generator in the interval $(\kappa, j_{E^*}(g)(\kappa))$. Such a situation, however, cannot occur if $g(\alpha) \leq \alpha^+$ for all α , as is also demonstrated in the proof of Theorem 0.2.

Remark 4.1. We conjecture that the requirement that $g(\alpha)$ lie below the least inaccessible above α can be relaxed by adopting a spaced nonstationary support.

Recall that the restriction on $g(\alpha)$ was imposed to ensure that the Fusion Lemma 1.3 applies to $\mathbb{P}$. Dropping this restriction, while still retaining a usable version of the Fusion Lemma, seems to require “spacing out” the nonstationary support—that is, requiring the support to be nonstationary only at inaccessibles from a specified stationary set.

For example, if κ is sufficiently large, one can force with supports that are nonstationary at inaccessible cardinals which are limits of inaccessibles. In this setting, the quotients $\mathbb{P} \setminus (\alpha + 1)$ (for $\alpha < \kappa$) exhibit stronger closure properties, making it possible to establish a version of the Fusion Lemma. With the bound on $g(\alpha)$ taken to be the least inaccessible limit of inaccessibles above α , Theorem 4.1 should then apply to a broader class of extenders. We will not pursue this direction further in the present paper.

Lemma 4.2. Assume that E is a (κ, λ) -extender in V whose generators all lie below $j_E(g)(\kappa)$. Let $G \subseteq \mathbb{R}$ be generic over V . Then in $V[G]$ there are η nonequivalent (κ, λ) -extendents $\langle E_\eta^*: \eta < j_E(g)(\kappa) \rangle \in V[G]$, all with the same ultrapower model, such that each $j_{E_\eta^*}$ is a lift of j_E .

Proof. Write $G = g' * G'$, where $g' \subseteq \text{Cohen}(\omega)$ is generic over V , and $G' \subseteq \mathbb{P}$ is generic over $V[g']$. The extender E naturally and uniquely lifts to an extender $E' \in V[g']$, namely the (κ, λ) -extender derived from j' , the standard lift of j to an embedding from $V[g']$ into $M_E[g']$. From this point onward, we work over $V[g']$ and aim to show that E' lifts in (at least) η distinct ways.

It suffices to prove that over any ground model V , whenever $G \subseteq \mathbb{P}$ is generic over V and $E \in V$ is a (κ, λ) -extender, there are at least $j_E(g)(\kappa)$ nonequivalent (κ, λ) -extendents $\langle E_\eta^*: \eta < j_E(g)(\kappa) \rangle$, all of them with the same ultrapower, such that each $j_{E_\eta^*}$ lifts j_E .

Thus, we change the settings to the ones mentioned in the previous paragraph. Fix $\eta < j_E(g)(\kappa)$. Define, in $V[G]$,

$$H_\eta = \{q \in j_E(\mathbb{P}) : \exists p \in G \ (q \subseteq j_E(p) \cup \{\langle \kappa, \eta \rangle\})\}.$$

We will argue that H_η is $j_E(\mathbb{P})$ -generic over V . The crux of the argument lies in proving that H_η meets every dense open set of M_E .

Fix a dense open subset $D \in M_E$ of $j_E(\mathbb{P})$. Let $a \in [j_E(g)(\kappa)]^{<\omega}$ be a set of generators and $e: [\kappa]^{|\alpha|} \rightarrow V$ be a function such that $D = j_E(e)(a)$. Define, for every $\alpha < \kappa$,

$$d(\alpha) = \{q \in \mathbb{P}: q \restriction \alpha+1 \Vdash \forall b \in [g(\alpha)]^{|\alpha|} \exists r \in \dot{G} \restriction \alpha+1 (r \frown (q \setminus (\alpha+1))) \in e(b)\}.$$

The fact that $g(\alpha)$ and $|\mathbb{P}_{\alpha+1}| = \max\{\alpha^+, g(\alpha)\}$ are both strictly below the least inaccessible above α , which is the amount of closure of $\mathbb{P} \setminus \alpha+1$, implies that $d(\alpha) \subseteq \mathbb{P}$ is dense open. Thus, we can apply the Fusion Lemma 1.3 and find a condition $p \in G$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$,

$$\{q \in \mathbb{P}_{\alpha+1}: q \frown (p \setminus \alpha+1) \in d(\alpha)\}$$

is dense open above $p \restriction \alpha+1$. In particular, for every such $\alpha \in C$,

$$p \restriction \alpha+1 \Vdash \forall b \in [g(\alpha)]^{|\alpha|} \exists r \in \dot{G} \restriction \alpha+1 (r \frown (p \setminus (\alpha+1))) \in e(b).$$

Since $C \subseteq \kappa$ is a club, $\kappa \in j_E(C)$, and thus

$$p \frown 0_{\mathbb{Q}_\kappa} \Vdash \exists r \in j_U(\dot{G}) \restriction \kappa+1 (r \frown (j_E(p) \setminus \kappa+1) \in D)$$

where we used the fact that $\bigcap_{b \in [j_E(g)(\kappa)]^{<\omega}} j_E(e)(b) \subseteq E$. Since G is $j_E(\mathbb{P}) \restriction \kappa = \mathbb{P}$ -generic over M_E , and η is $j_E(\langle \mathbb{Q}_\alpha: \alpha < \kappa \rangle)(\kappa)$ -generic over $M_E[G]$, we can extend p inside G and achieve $j_E(p) \cup \{\langle \kappa, \eta \rangle\} \in E \cap H_\eta$.

This proves that each H_η is $j_E(\mathbb{P})$ -generic over M_E . Since $j_E[G] \subseteq H_\eta$, j_E lifts to an embedding $j_{E,\eta}^*: V[G] \rightarrow M_E[H_\eta]$. By standard arguments, $j_{E,\eta}^*$ is the ultrapower embedding of the (κ, λ) -extender E_η^* derived from it. The extenders $\langle E_\eta^*: \eta < j_E(g)(\kappa) \rangle$ are nonequivalent, since their elementary embeddings disagree on their value at G . Nevertheless, for every $\eta \neq \eta'$ below $j_E(g)(\kappa)$, $M_E[H_\eta] = M_E[H_{\eta'}]$, namely $E_\eta^*, E_{\eta'}^*$ have the same ultrapower model. Indeed, $H_{\eta'}$ can be easily computed inside $M_E[H_\eta]$ by changing the generic value of H_η at coordinate κ from η to η' . $\square$

Remark 4.3. Note that η is not a generator of $j_{E,\eta}^* = j_{E_\eta^*}$, even if η was a generator of E , since $\eta = j_{E_\eta^*}(\cup G)(\kappa)$ in $V[G]$. In fact, this shows that $\mathbb{P}$ can be used to "kill" a generator of an extender $E \in V$.

This implies that normal measures on κ in the generic extension via $\mathbb{P}$ might rise from non-normal ground model measures (even though they restrict to a ground model normal measure, their ultrapower embedding lifts the ultrapower embedding of a non-normal measure). See Theorem 4.4 for more details on the characterization of normal measures in the new settings.

Theorem 4.2. Assume GCH. Let $G \subseteq \mathbb{R}$ be generic over V . Let E^* be a (κ, λ) -extender of $V[G]$ whose generators are all below $j_{E^*}(g)(\kappa)$ and assume that $\lambda \geq j_{E^*}(g)(\kappa)$. Then there exists a (κ, λ) -extender $E \in V$ and $\eta < j_E(g)(\kappa)$ such that E^* is equivalent to E_η^* .

Proof. Denote $M[H] \simeq \text{Ult}(V[G], E^*)$. By Hamkins' Gap forcing theorem, $j_{E^*} \restriction V: V \rightarrow M$ is definable in V . Let E be the (κ, λ) -extender derived from $j_{E^*} \restriction V$ in V .

We argue first that $j_E = j_{E^*} \restriction V$ and $M = M_E$. Define $k: M_E \rightarrow M$ such that, for every $a \in [\lambda]^{<\omega}$ and $f: [\kappa]^{|\alpha|} \rightarrow V$ in V , $k(j_E(f)(a)) = j_{E^*}(f)(a)$. It suffices to prove that k is the identity map.

Claim 4.4. *k is elementary and $j_{E^*} \restriction V = k \circ j_E$.*

Proof. Let $a \in [\lambda]^{<\omega}$ and assume that $f_0, f_1: [\kappa]^{|\alpha|} \rightarrow V$ are functions in V . Then:

$$\begin{aligned} j_E(f_0)(a) = j_E(f_1)(a) &\iff \{\vec{v} \in [\kappa]^{|\alpha|}: f_0(\vec{v}) = f_1(\vec{v})\} \in E_a \\ &\iff a \in j_{E^*} \left(\{\vec{v} \in [\kappa]^{|\alpha|}: f_0(\vec{v}) = f_1(\vec{v})\} \right) \\ &\iff j_{E^*}(f_0)(a) = j_{E^*}(f_1)(a). \end{aligned}$$

Building on this argument, full elementarity of k can be shown. $\square$

Note that $\text{crit}(k) \geq \lambda$ since every generator below λ is mapped to itself. In fact, the critical point of k is the least generator (if such exists) of $j_{E^*} \restriction V$ which is greater or equal λ . We argue that no such generator exists. Fix an ordinal $\zeta \geq \lambda$, and write, in $V[G]$, $\zeta = j_{E^*}(f)(a)$ for some $a \in [j_E(g)(\kappa)]^{<\omega}$ and $f: [\kappa]^{|\alpha|} \rightarrow \text{Ord}$ in $V[G]$. We would like to prove here a strengthening of Corollary 1.4:

Claim 4.5. *There exists a partial function $F \in V$ and a club $C \subseteq \kappa$ such that, for every $\alpha \in C$ and $\vec{v} \in [g(\alpha)]^{<\omega}$,*

$$f(\vec{v}) \in F(\alpha, \vec{v}) \text{ and } |F(\alpha, \vec{v})| \leq \max\{\alpha^+, g(\alpha)\}.$$

Proof. For every $\alpha < \kappa$, define

$$d(\alpha) = \{q \in \mathbb{P}: q \restriction \alpha + 1 \Vdash \forall \vec{v} \in [g(\alpha)]^{<\omega} \exists \gamma_\nu \in \text{Ord} (q \restriction \alpha + 1 \Vdash \dot{f}(\vec{v}) = \dot{\gamma}_\nu)\}.$$

Each $d(\alpha)$ is dense-open, since $\mathbb{P} \restriction \alpha + 1$ is more than $g(\alpha)$ -closed. By the Fusion Lemma 1.3 there exists $p \in G$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$ and $\vec{v} \in [g(\alpha)]^{<\omega}$, there exists a $\mathbb{P}_{\alpha+1}$ -name $\dot{\gamma}_\vec{v}^\alpha$ such that $p \Vdash \dot{f}(\vec{v}) = \dot{\gamma}_\vec{v}^\alpha$. Now, for every such $\alpha \in C$ and $\vec{v} \in [g(\alpha)]^{<\omega}$, let

$$F(\alpha, \vec{v}) = \{\gamma \in \text{Ord}: \exists r \geq p \restriction \alpha + 1 (r \Vdash \dot{\gamma}_\vec{v}^\alpha = \dot{\gamma})\}$$

and note that $|F(\alpha, \vec{v})| \leq |\mathbb{P}_{\alpha+1}| = \max\{g(\alpha), \alpha^+\}$ due to GCH and the fact that $|\mathbb{Q}_\alpha| = g(\alpha)$. $\square$

Now we are prepared to complete the proof that k is the identity map. Let C, F be as in the claim. Since C is a club in κ , $\kappa \in j_{E^*}(C)$. Since $a \in [\lambda]^{<\omega}$, we have that

$$\zeta = j_{E^*}(f)(a) \in (j_{E^*} \restriction V)(F)(\kappa, a).$$

Since we initially picked $\zeta \geq \lambda \geq j_{E^*}(g)(\kappa)$, and κ, a are all below $j_E(g)(\kappa)$, the above equality shows that ζ is not a generator of $j_{E^*} \restriction V$. As a corollary, we deduce that k is the identity map, and $j_{E^*} \restriction V = j_E$.

To sum up the situation so far, we have proved that $M = M_E$ and $j_{E^*} \restriction V = j_E$. Write $G = g' * G'$ where $g' \subseteq \text{Cohen}(\omega)$ is generic over V , and $G' \subseteq \mathbb{P}$ is generic over $V[g]$. Let $H = j_{E^*}(G)$ and write $H = g' * H'$, where H' is $j_{E^*}(\mathbb{P})$ -generic over $M[g']$. Note that $j_E[G] \subseteq H$ since j_{E^*} lifts j_E .

Denote $\eta = j_{E^*}(\cup G')(\kappa) < j_{E^*}(g)(\kappa)$. Again since j_{E^*} lifts j_E , we deduce that $\eta < j_E(g)(\kappa)$, and thus the lift $E_\eta^* \in V[G]$ is defined.

Let $j': V[g'] \rightarrow M[g']$ be the natural lift of j . Following the proof of Lemma 4.2, the set

$$H'_\eta = \{q \in j'(\mathbb{P}) : \exists p \in G' (q \subseteq j'(p) \cup \{\langle \kappa, \eta \rangle\})\}$$

is $j'(\mathbb{P})$ -generic over $M[g']$. Since $\eta = j_{E^*}(\cup G')(\kappa) = (\cup H')(\kappa)$ and $j'[G'] \subseteq H'$, we obtain that $H'_\eta \subseteq H'$, namely $H'_\eta = H'$. Thus $\text{Ult}(V[G], E^*) = M[H] = M_E[g' * H'_\eta]$ and $j_{E^*} = j_{E_\eta^*}$. It follows that E^* and E_η^* are equivalent, as desired. $\square$

We can now conclude the proofs of the main results of this section.

Proof of Theorem 4.1. The first clause of Theorem 4.1 follows from Lemma 4.2, and the second clause follows from Theorem 4.2. $\square$

Proof of Theorem 0.3. Fix $E \in V$ a (κ, κ^{++}) extender witnessing that κ is κ^{++} -strong. For every $\alpha < \kappa$, let $g(\alpha) = \alpha^{++}$. Let $G \subseteq \mathbb{R}$ be generic over V . By Theorem 4.1, E lifts in exactly $j_E(g)(\kappa) = (\kappa^{++})^{M_E} = \kappa^{++}$ ways to an extender of $V[G]$, and a standard argument (see, for instance, [8, Proposition 8.4]) shows that each of them witnesses that κ is $(\kappa + 2)$ -strong in $V[G]$. On the other hand, each (κ, κ^{++}) -extender $E^* \in V[G]$ witnessing $(\kappa + 2)$ -strongness is, again by Theorem 4.1, such a lift of some ground model (κ, κ^{++}) -extender $E \in V$. By Hamkins' Gap-forcing theorem 2.1, $M_{E^*}^{V[G]} \cap V = M_E$, and $V_{\kappa+2} \subseteq M_{E^*}^{V[G]} \cap V$. It follows that $V_{\kappa+2} \subseteq M_E$ and E witnesses $(\kappa + 2)$ -strongness in V . $\square$

Goldberg observed that even though there are possibly κ^{+3} -many (κ, κ^{++}) -extenders (in the GCH context), only κ^{++} of them may have the same ultrapower. In this sense, Theorem 0.5 is optimal.

Theorem 4.3 (Goldberg). *Denote by β the number of ultrafilters on κ . Let E be a (κ, λ) -extender. Then, up to equivalence, there are at most $\max\{\beta, \lambda\}$ extenders with the same ultrapower as E .*

Proof. Following [12], we say that an elementary embedding $i: V \rightarrow N$ is almost an ultrapower embedding if for every $X \subseteq N$ there exists $a \in N$ such that $X \subseteq \mathcal{H}^N(i[V] \cup \{a\})$ (here and below, $\mathcal{H}$ denotes the Skolem hull).

Claim 4.6. *Let $j: V \rightarrow M$ be an extender embedding, $i: V \rightarrow N$ almost an ultrapower embedding, and $k: N \rightarrow M$ an elementary embedding such that $j = k \circ i$. Then i is an ultrapower embedding; that is, there exists $a \in N$ such that $N = \mathcal{H}^N(i[V] \cup \{a\})$.*

Proof. Since j is an extender embedding, there exists a set $A \in V$ such that $j(A)$ covers the set of (finite sequences of) generators of j . In particular, $M = \mathcal{H}^M(j[V] \cup j(A))$. It follows that $N = \mathcal{H}^N(j[V] \cup i(A))$: for every element $x \in N$, there exists a function $f \in V$ such that

$$M \models \exists a \in j(A) (k(x) = j(f)(a)),$$

where we view $k(x)$, $j(A) = k(i(A))$ and $j(f) = k(i(f))$ as parameters. By elementarity of k ,

$$N \models \exists a \in i(A) (x = i(f)(a)),$$

So $x \in \mathcal{H}^N(j[V] \cup i(A))$.

Since i is almost an ultrapower embedding, for every ordinal ξ there exists $\alpha_\xi \in N$ such that $(V_\xi)^N \subseteq \mathcal{H}^N(j[V] \cup \{\alpha_\xi\})$. Each such α_ξ belongs to $\mathcal{H}^N(j[V] \cup \{a_\xi\})$ for some $a_\xi \in i(A)$. Thus, there exists $a \in i(A)$ such that for unboundedly many ordinals ξ , $\alpha_\xi \in \mathcal{H}^N(j[V] \cup \{a\})$. Thus, $N \subseteq \mathcal{H}^N(j[V] \cup \{a\})$. $\square$

Having this in mind, fix an extender E as in the formulation of the Lemma. By [12, Theorem 3.7], for every extender F with the same ultrapower as E there exists an inner model N (actually, $N = \mathcal{H}^M(j_E[V] \cup j_F[V])$) and elementary embeddings $i_0: V \rightarrow N$, $i_1: V \rightarrow N$ and $k: N \rightarrow M_E$ such that $j_E = k \circ i_0$, $j_F = k \circ i_1$, and i_0, i_1 are almost ultrapower embeddings. By the above claim, since i_0, i_1 factor into extender embeddings, both are ultrapower embeddings with respect to ultrafilters Z_0, Z_1 on κ , respectively.

We argue that k is uniquely determined from E , a function $f: \kappa \rightarrow \kappa$ and some $a \in [\lambda]^{<\omega}$. Indeed, let $a \in [\lambda]^{<\omega}$ and $f \in V$ be such that $k([Id]_{Z_0}) = j_E(f)(a)$. Then $Z_0 = f_*(E_a) = \{X \subseteq \kappa: f^{-1}[X] \in E_a\}$, and k maps each element of the form $i_0(g)([Id]_{Z_0})$ to $j_E(g)(j_E(f)(a))$. Thus k is uniquely determined from E , f and a .

Since $j_F = k \circ j_{Z_1}$, F is uniquely determined from Z_1 , E , f and $a \in [\lambda]^{<\omega}$. The set of all such parameters has size $\max\{\beta, \lambda\}$. $\square$

We finish this section with a complete classification of normal measures in $V[G]$ for $G \subseteq \mathbb{R}$ generic over V . By Remark 4.3, normal measures $W \in V[G]$ may rise-up from non-normal measures E of V which have an additional single generator above κ . In this case, the forcing $\mathbb{P}$ might "kill" this additional generator, and E lifts to a normal measure which differs from those described in Theorem 0.2.

We summarize this in the following theorem.

Theorem 4.4. *Assume GCH. Let $G \subseteq \mathbb{R}$ be generic over V . Suppose that $W \in V[G]$ is a normal measure on κ . Then there exists a $(\kappa, j_W(g)(\kappa))$ -extender E in V such that $W = E_\eta^*$ for some $\eta < j_E(g)(\kappa)$. Furthermore, E is equivalent to a κ -complete ultrafilter on κ , which is the one derived from j_E using (κ, η) as a seed.*

Proof. As usual, write $G = g' * G'$, where $g' \subseteq \text{Cohen}(\omega)$ is generic over V , and $G' \subseteq \mathbb{P}$ is generic over $V[g']$.

Denote $\lambda = j_W(g)(\kappa) < j_W(\kappa)$. Let E^* be the (κ, λ) -extender derived from j_W . Note that $j_W = j_{E^*}$, since the only generator of W is κ , and it is below $j_W(g)(\kappa)$. By Theorem 4.2, we deduce that $j_W = j_{E_\eta^*}$, for some (κ, λ) -extender E in V and $\eta < j_E(g)(\kappa)$. By the analysis in the proof of Theorem 4.2, $\eta = j_W(\cup G')(\kappa)$.

It remains to prove that every $x \in M_E = \text{Ult}(V, E)$ has the form $j_E(\kappa, \eta)$, and thus E is equivalent to the measure derived from j_E using (κ, η) as a seed.¹¹

Fix $\zeta \neq \kappa, \eta$. Since W is normal, there exists $f \in V[G]$ such that $\zeta = j_W(f)(\kappa)$.

Work in $V[g']$. By the standard Fusion argument from the proof of Corollary 1.4, there exists $p \in G'$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$,

$$p \restriction \alpha + 1 \Vdash \exists \gamma \in \text{Ord} \left(p \restriction (\alpha + 1) \Vdash \dot{f}(\alpha) = \dot{\gamma} \right).$$

Let $j': V[g'] \rightarrow M_E[g']$ be the natural lift of j_E . Since $\kappa \in j'(C)$, we have in $M_E[g']$ that

$$p \restriction 0_{\mathbb{Q}_\kappa} \Vdash \exists \gamma \in \text{Ord} \left(j'(p) \restriction (\kappa + 1) \Vdash j'(\dot{f})(\kappa) = \dot{\gamma} \right).$$

We can use the fact that $\eta < j_E(\tau)(\kappa)$ is a possible generic for $\mathbb{Q}_\kappa$, to deduce

$$p \Vdash \exists \gamma \in \text{Ord} \left(\{(\kappa, \eta)\} \cup j'(p) \restriction (\kappa + 1) \Vdash j'(\dot{f})(\kappa) = \dot{\gamma} \right).$$

Overall, we proved that, in $M_E[g]$, there exists a $j'(\mathbb{P}) \restriction \kappa = \mathbb{P}$ -name $\dot{\gamma}$ for an ordinal, such that

$$(j'(p) \cup \{(\kappa, \eta)\}) \Vdash j'(\dot{f})(\kappa) = \dot{\gamma}.$$

We can assume that $\dot{\gamma}$ is the least $\mathbb{P}$ -name as above, where, by "least", we mean here that we fixed in advance a well order $\mathcal{W}$ of V_κ , which induces a well order $\mathcal{W}'$ of $(V_\kappa)^{V[g']}$, and chose $\dot{\gamma}$ least with respect to $j'(\mathcal{W}')$. It follows that $\dot{\gamma}$ has the form $j'(h)(\kappa, \eta)$ for some function $h \in V[g']$.

Let $A \in M_E$ be the set of all possible values of $\dot{\gamma}$, namely

$$A = \{\beta \in \text{Ord} : \exists q \in \mathbb{R} (q \Vdash \dot{\gamma} = \check{\beta})\}.$$

Then A has the form $j(h^*)(\kappa, \eta)$ for some function $h^* \in V$ (note that $\mathbb{R} = j_E(\mathbb{R}) \restriction \kappa$ in the above definition of A), and $|A| \leq \kappa^+$ since $|\mathbb{R}| = \kappa^+$.

Recall that $p \in G'$ and $\eta = j_W(\cup G')(\kappa)$. Thus, in $\text{Ult}(V[G], W) = M_E[g' * j_W(G')]$, we have $\zeta = j_W(f)(\kappa) \in A$.

In particular, for some ordinal $\xi < \kappa^+$, ζ is the ξ -th element of A with respect to the least well-order of A of order type $\leq \kappa^+$ (again, the well order is

¹¹possibly, E is already equivalent to a normal measure, and then κ is the single generator of j_E and W is really a lift of a ground-model normal measure. This happens, for instance, if $j_E(g)(\kappa) \leq \kappa^+$.

chosen minimal with respect to $j_E(\mathcal{W})$). Letting f_ξ be the canonical function that represents ξ , we obtain that, in M_E , ζ is the $j_E(f_\xi)(\kappa)$ -th element of $j_E(h^*)(\kappa, \eta)$, with respect to the $j_E(\mathcal{W})$ -least well order of $j_E(h^*)(\kappa, \eta)$ of order type κ^+ .

All of this shows that $\zeta = j_E(h')(\kappa, \eta)$ for some function $h' \in V$. Since ζ was an arbitrary ordinal outside of $\{\kappa, \eta\}$, we deduce that E is equivalent to the κ -complete ultrafilter derived from j_E using (κ, η) as seeds. $\square$

5 Open questions and concluding remarks

In this final section, we discuss some open questions.

Question 5.1. *Is it consistent that GCH holds, there are κ^{++} many normal measures on κ , but only a single normal ultrapower?*

The same proof technique of Theorem 0.2 does not appear to apply here (see Remark 2.2), but we still conjecture that the answer is positive.

Note that a positive answer to Question 5.1 would exhibit behavior somewhat different from that of extenders. For example, assuming GCH, there are at most κ^{++} many (κ, κ^{++}) -extenders with the same ultrapower (by Theorem 4.3), even though the total number of (κ, κ^{++}) -extenders can be κ^{+3} .

Next, recall that the Friedman–Magidor theorem generalizes well to the context of violation of GCH on a measurable.

Theorem 5.1 (Friedman and Magidor, [9]). *Let $V = L[\bar{E}]$ be the minimal extender model that contains a cardinal κ which is $(\kappa + 2)$ -strong. Then for every $\tau < \kappa^{++}$, there exists a cardinal preserving forcing extension in which κ is measurable, $2^\kappa = \kappa^{++}$ and there are exactly τ normal measures on κ .*

This motivates the question whether the methods developed in this paper can yield similar results.

Question 5.2. *Assume that κ is $(\kappa + 2)$ -strong in an arbitrary ZFC+GCH model V . Fix $\tau < \kappa^{++}$. Is there a cardinal preserving forcing extension $V[G]$ such that:*

1. κ is measurable and $2^\kappa = \kappa^{++}$ in $V[G]$.
2. For every extender $E \in V$ witnessing that κ is $(\kappa + 2)$ -strong, there are τ normal measures on κ in $V[G]$, such that for each of them, say U , $j_U^{V[G]}$ lifts j_E .
3. Every normal measure on κ arises as a lift of some strong extender E as in the previous clause.

In [5], it was shown that the Ultrapower Axiom is consistent with the failure of GCH at a measurable cardinal, via the construction of an $L[U]$ -like model: a model in which κ is the unique measurable cardinal, carries a single normal measure U , and every other σ -complete ultrafilter is equivalent to U^n for some $n < \omega$. This result motivates the following question:

Question 5.3. *Is the Weak Ultrapower Axiom and the negation of the Ultrapower Axiom consistent with the failure of GCH on a measurable?*

A natural approach to Question 5.3 is to force with the splitting forcing over the $L[U]$ -like model described above and analyze all possible lifts of U^n , as in Section 3. Indeed, the proof of Theorem 0.4 confirms that this construction yields a model of the Weak UA. However, applying the splitting forcing to a ground model in which GCH already fails at κ may collapse κ^{++} and inadvertently restore GCH. In fact, this is precisely what happens when forcing with the splitting forcing over the $L[U]$ -like model, as the following claim shows:

Claim 5.4. *Let $\mathbb{P}$ be the splitting forcing, and assume that $2^\kappa = \kappa^{++}$. Suppose there exists a collection $\mathcal{C}$ of clubs in κ , such that $|\mathcal{C}| = \kappa^+$ and every club $D \subseteq \kappa$ contains a club from $\mathcal{C}$.¹² Then the splitting forcing $\mathbb{P}$ collapses κ^{++} .¹³*

Proof. Let $\mathbb{P}$ be the splitting forcing, and let $G \subseteq \mathbb{P}$ be generic over V . We claim that in $V[G]$ there exists a surjection from $\mathcal{C}$ to $(\mathcal{P}(\kappa))^V$.

We say that for $C \in \mathcal{C}$ and $A \subseteq \kappa$ in V , the generic filter G codes A via C if, for every $i < \kappa$,

$$i \in A \iff (\text{UG})(C_i^+) = 1$$

where C_i^+ denotes the i -th successor point of C .

It suffices to prove that for every $A \subseteq \kappa$ in V there exists $C \in \mathcal{C}$ such that G codes A via C .

Indeed, fix $A \subseteq \kappa$ in V and a condition $p \in \mathbb{P}$. By assumption, there is a club $C \in \mathcal{C}$ disjoint from $\text{supp}(p)$. The sequence $\langle C_i^+ : i < \kappa \rangle$ is nowhere stationary, so we may extend p to a condition $q \in \mathbb{P}$ as follows:

$$q = p \cup \{ \langle C_i^+, 1 \rangle : i \in A \} \cup \{ \langle C_i^+, 0 \rangle : i \notin A \}.$$

By density, there exists $q \in G$ forcing that $\dot{G}$ codes A via some $C \in \mathcal{C}$. □

Claim 5.4 naturally leads to the following question.

Question 5.5. *Is there a variant of the splitting forcing that preserves κ^{++} even when $2^\kappa = \kappa^{++}$ in V ?*

Finally, the following central question, raised by Friedman and Magidor in [9], has remained open ever since:

Question 5.6. *Is there a Friedman–Magidor theorem for fine, normal measures on $\mathcal{P}_\kappa(\lambda)$ for $\lambda > \kappa$?*

¹²For instance, this holds if V is the generic extension of a GCH model V_0 , such that every $f: \kappa \rightarrow \kappa$ in V is dominated by some $g: \kappa \rightarrow \kappa$ in V_0 .

¹³A similar argument shows that any other reasonable nonstationary support product or iteration collapses κ^{++} under the same assumptions.

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